Priyanshu Singh

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Education

B.Tech in Electronics and Communication Engineering **2022 – 2026**
Vellore Institute of Technology, Vellore *CGPA: 8.39/10*

Class XII (AISSCE): 85.2% | **Class X (AISSE): 95.4%** **2022 / 2020**
Delhi Public School, Siliguri

Skills

Languages: TypeScript, JavaScript, Python, SQL, C++, Bash
Backend & Frontend: Node.js, Express.js, Fastify, FastAPI, React.js, Tailwind CSS, BullMQ, SQLAlchemy
Databases: PostgreSQL (indexing, query optimization), pgvector, Redis, MongoDB
AI & LLMs: Tool-Calling Agents, RAG, Embeddings, Hybrid Search, LLM Evaluation, Gemini API, Claude Code
Cloud, DevOps & Testing: AWS (EC2, S3), Docker, GitHub Actions, pytest, Vitest, Testcontainers, Playwright, k6
Concepts: Idempotency, At-Least-Once Delivery, Event-Driven Architecture, WebSockets, Multi-Tenancy, DSA

Projects

Groundwork: AI Document Q&A Agent with Citations GitHub | Live Demo

- Built a multi-tenant **tool-calling AI agent** (FastAPI, Gemini, pgvector) that answered **90/90** test questions on a synthetic set of invoices and contracts, citing the source page or the rows behind each total.
- Raised **recall@5 from 81.7% to 93.3%** with **hybrid search** (pgvector + Postgres full-text, rank fusion); a cross-encoder reranker scored 78.3% at about 1.4 s, so it stays disabled.
- Lifted multi-hop accuracy from **6/20 to 20/20** with validated contract tables queried through **parameterised SQL tools**; the model never writes SQL.
- Deployed on **AWS** (EC2, S3, SSM; no SSH or stored keys) with a tested backup restore; **396 backend tests (93% coverage)** including tenant-isolation and prompt-injection tests.
- **Tech Stack:** Python, FastAPI, PostgreSQL, pgvector, Gemini API, React, TypeScript, Docker, AWS, pytest.

Real-Time Notification Service GitHub | Video Demo

- Architected a multi-tenant **event-driven** notification service (Fastify, PostgreSQL, Redis, **BullMQ**) with **22 REST/WebSocket operations**, email and real-time in-app delivery.
- Engineered **at-least-once delivery** with **idempotency keys**, state-machine transitions and backoff retries; SIGKILLing a worker with 20 in flight left **0 lost, 0 duplicated**.
- Load-tested with **k6**: **18,000 notifications/min, 0 loss, 24 ms p95** (laptop, mock email); **86 tests (79.84% coverage)** with Testcontainers in GitHub Actions CI.
- **Tech Stack:** TypeScript, Node.js, Fastify, PostgreSQL, Redis, BullMQ, WebSockets, React, Docker, Vitest, k6.

Distraction Tracker GitHub | Video Demo

- Built a **Chrome MV3 extension**, **React/TypeScript** dashboard and **Express/PostgreSQL** API solo, with retry-safe batched uploads de-duplicated on client UUIDs.
- Cut analytics query time **33x (396 ms to 11.95 ms**, summed p50 of 7 queries, **1M rows)** via a **composite index** plus time-zone-aware range predicates.
- Hardened **JWT** auth (httpOnly cookies, server-side revocation, scoped tokens); **1,182 passing tests** (95.34% server coverage) caught 23/23 deliberate auth mutations.
- **Tech Stack:** React, TypeScript, Tailwind CSS, Node.js, Express.js, PostgreSQL, Manifest V3, Docker, Vitest.

Achievements

- **LeetCode Knight** (contest rating **1915**, Top 4.26%); best ranks **193** (Biweekly 171) and **465** (Biweekly 167). [Link]
- Solved **650+ DSA problems** across LeetCode, GeeksforGeeks and other platforms. [Link]